11.3 The Dot Product

Given two vectors $\mathbf{u} =< u_1, u_2, u_3 >$ and $\mathbf{v} =< v_1, v_2, v_3 >$ in $\mathbb{R}^3$, the **dot product** of the vectors $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u} \cdot \mathbf{v}$, is defined by

$$\underset{\text{vectors}}{\mathbf{u} \cdot \mathbf{v}} = u_1v_1 + u_2v_2 + u_3v_3. \quad \Rightarrow \text{output is a scalar}$$

In the two dimensional space, if $\mathbf{u} =< u_1, u_2 >$ and $\mathbf{v} =< v_1, v_2 >$, then

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2.$$

Example:

1. $\mathbf{u} =< 1, 2, 3 >, \mathbf{v} =< 3, 2, -1 >$

   $\underline{u} \cdot \underline{v} = u_1v_1 + u_2v_2 + u_3v_3 = 1 \cdot 3 + 2 \cdot 2 + 3 \cdot (-1) = \underline{4}$

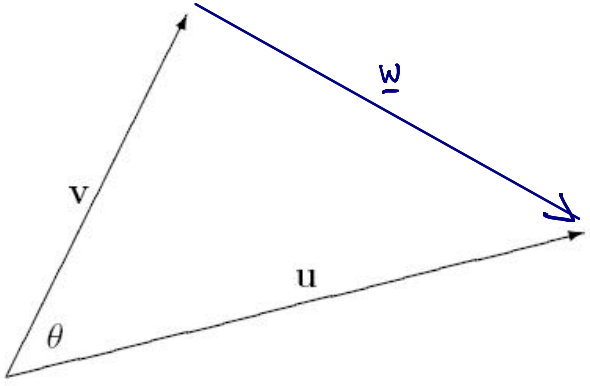
2. $\mathbf{u} = \frac{1}{2}\mathbf{i} + 3\mathbf{j} + \mathbf{k}, \mathbf{v} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

   $\underline{u} = \langle \frac{1}{2}, 3, 1 \rangle$, $\underline{v} = \langle 4, -1, 2 \rangle$ $\quad \underline{u} \cdot \underline{v} = \frac{1}{2}(4) + 3(-1) + 1(2)$

3. $\mathbf{u} =< \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}} >, \mathbf{v} =< \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{3}} >$ $\quad = \underline{1}$

   $\underline{u} \cdot \underline{v} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$

Angle Between Vectors



$$\mathbf{u} = < u_1, u_2, u_3 >$$
$$\mathbf{v} = < v_1, v_2, v_3 >$$

$\underline{u} = \underline{v} + \underline{w}$

$\Rightarrow \underline{w} = \underline{u} - \underline{v}$

$= < u_1 - v_1 , u_2 - v_2 , u_3 - v_3 >$

Recall the cosine law : $|\underline{w}|^2 = |\underline{u}|^2 + |\underline{v}|^2 - 2|\underline{u}||\underline{v}|\cos\theta$.

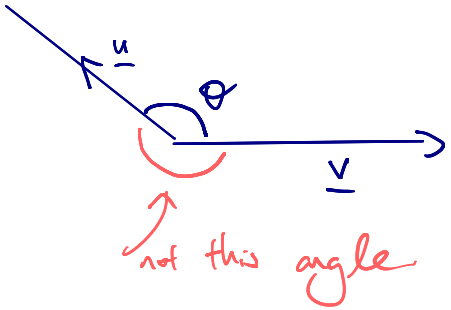
$$|\underline{w}|^2 = (u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2$$
$$= (u_1^2 + u_2^2 + u_3^2) + (v_1^2 + v_2^2 + v_3^2) - 2(u_1v_1 + u_2v_2 + u_3v_3)$$
$$= |\underline{u}|^2 \quad |\underline{v}|^2 \quad \underline{u}\cdot\underline{v}$$
$$= |\underline{u}|^2 + |\underline{v}|^2 - 2\,\underline{u}\cdot\underline{v}$$

$$\Rightarrow \quad \underline{u}\cdot\underline{v} = |\underline{u}||\underline{v}|\cos\theta \Rightarrow \cos\theta = \frac{\underline{u}\cdot\underline{v}}{|\underline{u}||\underline{v}|}$$

Theorem: The angle θ between two vectors $\mathbf{u}$ and $\mathbf{v}$ is given by

$$\theta = \cos^{-1}\left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right).$$

Remark: We take θ in the range $0 \leq \theta \leq \pi$.



Remark: Dot products are also called *inner* or *scalar products* because the product results in a scalar, not a vector.

Example:

Find the angle between the vectors $\mathbf{u}=\sqrt{3}\mathbf{i}-5\mathbf{j}$ and $\mathbf{v}=\sqrt{3}\mathbf{i}+\mathbf{j}-2\mathbf{k}$.

$$\cos\theta = \frac{\underline{u}\cdot\underline{v}}{|\underline{u}||\underline{v}|}$$

$$\underline{u}\cdot\underline{v} = (\sqrt{3})(\sqrt{3}) - 5(1) = -2$$

$$|\underline{u}| = \sqrt{(\sqrt{3})^2 + (-5)^2} = \sqrt{28} = 2\sqrt{7}.$$

$$|\underline{v}| = \sqrt{(\sqrt{3})^2 + 1^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}.$$

$$\cos\theta = \frac{\underline{u}\cdot\underline{v}}{|\underline{u}||\underline{v}|} = \frac{-2}{2\sqrt{7}\,2\sqrt{2}} = -\frac{1}{2\sqrt{14}}$$

$$\Rightarrow \quad \theta = \cos^{-1}\left(-\frac{1}{2\sqrt{14}}\right).$$

Perpendicular (Orthogonal) Vectors

↳ the angle being $\frac{\pi}{2}$

Two nonzero vectors **u** and **v** are **orthogonal** (or **perpendicular**) if and only if **u•v**= 0.

"$\Rightarrow$" $\underline{u} \perp \underline{v}$, so $\theta = \frac{\pi}{2} \Rightarrow \cos\theta = 0$

$$\underline{u} \cdot \underline{v} = |\underline{u}|\,|\underline{v}| \cos\theta = 0$$

"$\Leftarrow$" $\underline{u} \cdot \underline{v} = |\underline{u}|\,|\underline{v}| \cos\theta = 0$

$\underline{u}$ & $\underline{v}$ non-zero $\Rightarrow |\underline{u}| \neq 0$, $|\underline{v}| \neq 0$

$\Rightarrow \cos\theta = 0$

$\Rightarrow \theta = \frac{\pi}{2} \Rightarrow \underline{u} \perp \underline{v}$

Example: Suppose $P(1, 1, -1)$, $Q(0, 3, -2)$, $S(k, 0, 2)$ are points in space. Given that $\overrightarrow{PQ}$ is orthogonal to $\overrightarrow{PS}$, find k.

$\overrightarrow{PQ} \perp \overrightarrow{PS}$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \langle 0, 3, -2 \rangle - \langle 1, 1, -1 \rangle$

$= \langle -1, 2, -1 \rangle$

$\overrightarrow{PS} = \langle k-1, -1, 3 \rangle$.

$\overrightarrow{PQ} \perp \overrightarrow{PS} \Rightarrow \overrightarrow{PQ} \cdot \overrightarrow{PS} = 0$

$0 = \overrightarrow{PQ} \cdot \overrightarrow{PS} = (-1)(k-1) + 2(-1) + (-1)(3)$

$= -k - 4$

$\Rightarrow \underline{k = -4}$

Dot Product Properties

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and let c be a scalar. Then,

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ ← commutative law
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ ← distributive law .
4. $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$
5. $\mathbf{0} \cdot \mathbf{u} = 0$

Example:

Suppose $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors such that $|\mathbf{u}| = |\mathbf{v}|$. Show that $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal.

We want to show that

$$(\underline{u} + \underline{v}) \cdot (\underline{u} - \underline{v}) = 0$$

$$(\underline{u} + \underline{v}) \cdot (\underline{u} - \underline{v}) = \underline{u} \cdot \underline{u} + \underline{v} \cdot \underline{u} - \underline{u} \cdot \underline{v} + \underline{v} \cdot \underline{v}$$

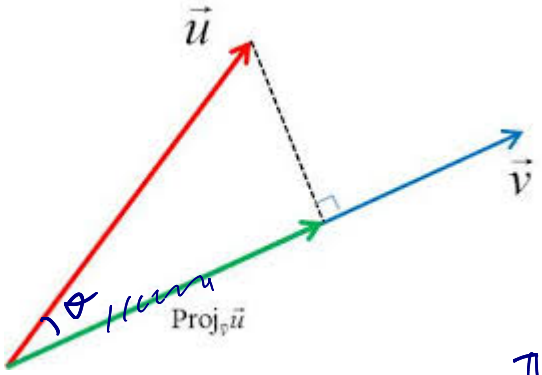
$= \underline{u} \cdot \underline{v}$ by commutative law

$$= |\underline{u}|^2 - |\underline{v}|^2$$

$$= 0 \quad (\text{since } |\underline{u}| = |\underline{v}|)$$

$\Rightarrow$ $\underline{u} + \underline{v}$ $\perp$ $\underline{u} - \underline{v}$

Vector Projections

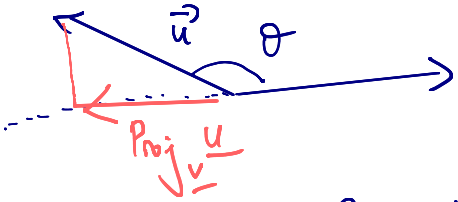


Drop the terminal point of $\mathbf{u}$ perpendicular to the line containing $\mathbf{v}$, we have

$$\text{Proj}_{\mathbf{v}}\mathbf{u}.$$

(vector projection of $\underline{u}$ onto $\underline{v}$)

The scalar component of $\text{Proj}_{\underline{v}}\underline{u}$ is $|\underline{u}|\cos\theta$, where θ is the angle between $\underline{u}$ & $\underline{v}$.



$$\text{Proj}_{\underline{v}}\underline{u} = (|\underline{u}|\cos\theta)\frac{\underline{v}}{|\underline{v}|}$$

$$\underline{u}\cdot\underline{v} = |\underline{u}||\underline{v}|\cos\theta$$

$$= \left(\frac{\underline{u}\cdot\underline{v}}{|\underline{v}|}\right)\frac{\underline{v}}{|\underline{v}|}$$

$$= \left(\frac{\underline{u}\cdot\underline{v}}{|\underline{v}|^2}\right)\underline{v}$$

Example: Find the vector projection of $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$

$$\text{Proj}_{\underline{v}}\,\underline{u} = \left(\frac{\underline{u}\cdot\underline{v}}{|\underline{v}|^2}\right)\underline{v}$$

$$\underline{u}\cdot\underline{v} = 6 - 6 - 4 = -4$$

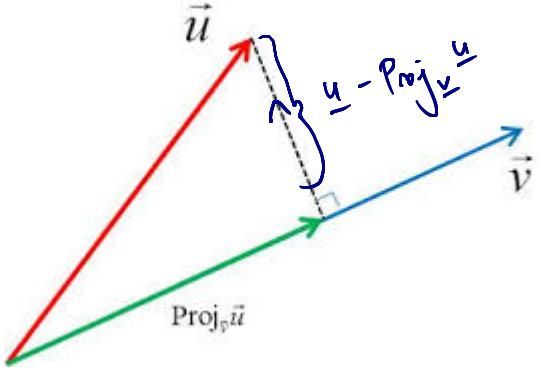
$$|\underline{v}|^2 = 1 + 4 + 4 = 9 \Rightarrow |\underline{v}| = 3$$

$$\Rightarrow \text{Proj}_{\underline{v}}\,\underline{u} = \frac{-4}{9}\langle 1, -2, -2\rangle$$

$$= -\frac{4}{9}\underline{i} + \frac{8}{9}\underline{j} + \frac{8}{9}\underline{k}$$

The scalar component of $\text{Proj}_{\underline{v}}\,\underline{u}$ is $\dfrac{\underline{u}\cdot\underline{v}}{|\underline{v}|} = -\dfrac{4}{3}$

Writing a Vector as a Sum of Orthogonal Vectors



Given $\underline{u} = \langle u_1, u_2, u_3 \rangle$

We can write

$$\underline{u} = u_1 \underline{i} + u_2 \underline{j} + u_3 \underline{k}$$

($\underline{i}, \underline{j}, \underline{k}$ are orthogonal vectors.)

Observe that $\text{Proj}_{\underline{v}} \underline{u}$ $\perp$ $(\underline{u} - \text{Proj}_{\underline{v}} \underline{u})$

So we can write $\underline{u}$ as a sum of two orthogonal vectors as follows.

$$\underline{u} = \underbrace{\text{Proj}_{\underline{v}} \underline{u}}_{\text{vector} \parallel \underline{v}} + \underbrace{(\underline{u} - \text{Proj}_{\underline{v}} \underline{u})}_{\text{vector } \perp \text{ to } \underline{v}}$$

Example:

A force $\mathbf{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ is applied to a spacecraft with velocity vector $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$. Express $\mathbf{F}$ as a sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

$$\underline{F} = \text{Proj}_{\underline{v}} \underline{F} + (\underline{F} - \text{Proj}_{\underline{v}} \underline{F})$$

$$\text{Proj}_{\underline{v}} \underline{F} = \frac{\underline{F} \cdot \underline{v}}{|\underline{v}|^2} \underline{v}$$

$$\underline{F} \cdot \underline{v} = 2(3) + 1(-1) = 5$$

$$|\underline{v}|^2 = 10$$

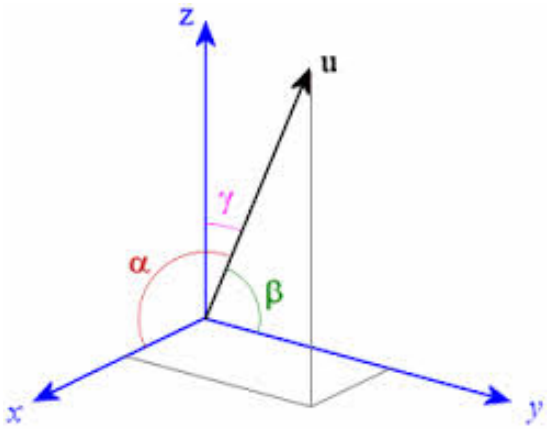
$$\Rightarrow \text{Proj}_{\underline{v}} \underline{F} = \frac{5}{10}(3\underline{i} - \underline{j}) = \frac{3}{2}\underline{i} - \frac{1}{2}\underline{j}$$

$$\underline{F} - \text{Proj}_{\underline{v}} \underline{F} = \frac{1}{2}\underline{i} + \frac{3}{2}\underline{j} - 3\underline{k}$$

$$\Rightarrow \underline{F} = \left\langle \frac{3}{2}, -\frac{1}{2}, 0 \right\rangle + \left\langle \frac{1}{2}, \frac{3}{2}, -3 \right\rangle$$

The Direction Cosines of Vectors in Space

Let $\mathbf{u} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, and let α, β, γ be the directed angles from the positive x-axis, y-axis, and z-axis, respectively, to the vector $\mathbf{u}$.



Then we have the following definitions.

- α, β, γ are called the *direction angles* of $\mathbf{u}$.
- (cos α, cos β, cos γ) are the *direction cosines* of $\mathbf{u}$.
- The ratios a: b: c are called the *direction ratios* of $\mathbf{u}$.

Theorem:

If α, β, γ are the direction angles of $\mathbf{u} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \neq \mathbf{0}$, then

1. $\cos\alpha = \frac{a}{|\mathbf{u}|}$, $\cos\beta = \frac{b}{|\mathbf{u}|}$ and $\cos\gamma = \frac{c}{|\mathbf{u}|}$.

2. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

Proof: ① α = angle between $\underline{u}$ & $\underline{i}$

$$\Rightarrow \cos\alpha = \frac{\underline{u}\cdot\underline{i}}{|\underline{u}||\underline{i}|} = \frac{a}{|\underline{u}|}$$

Similarly, $\cos\beta = \frac{b}{|\underline{u}|}$, $\cos\gamma = \frac{c}{|\underline{u}|}$

② $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{a^2}{|\underline{u}|^2} + \frac{b^2}{|\underline{u}|^2} + \frac{c^2}{|\underline{u}|^2} = \frac{a^2+b^2+c^2}{|\underline{u}|^2}$

$$= 1$$

Example: Find the direction angles of $\mathbf{v} = \mathbf{i} + \mathbf{j} + \sqrt{2}\mathbf{k}$.

$|\underline{v}| = \sqrt{1+1+2} = 2$

$\Rightarrow \cos\alpha = \frac{1}{2}$, $\cos\beta = \frac{1}{2}$, $\cos\gamma = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$.

$\Rightarrow \alpha = \frac{\pi}{3}$, $\beta = \frac{\pi}{3}$, $\gamma = \frac{\pi}{4}$

$\Rightarrow$ the direction angles of $\underline{v}$ are $\left(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{4}\right)$.

Remarks:

1. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma \neq 1 \Rightarrow$ does not exist vector with direction cosines $(\cos\alpha,\ \cos\beta,\ \cos\gamma)$.

2. $\mathbf{u}$ and $\mathbf{v}$ have the same direction cosines $\Leftrightarrow$ they have the same direction.

3. $\mathbf{u}$ and $\mathbf{v}$ have the same direction ratios $\Leftrightarrow$ they are parallel (i.e., they are in the same direction or in opposite direction)

4. Any vector $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \neq \mathbf{0}$ can be written as
$\mathbf{v} = |\mathbf{v}|(\cos\alpha\mathbf{i} + \cos\beta\mathbf{j} + \cos\gamma\mathbf{k})$.

5. The unit vector $\mathbf{u}$ in the direction $\mathbf{v}$ is given by
$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = (\cos\alpha\mathbf{i} + \cos\beta\mathbf{j} + \cos\gamma\mathbf{k})$.